MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING

STUDY GUIDE

G20

Agenda 01: Assessing the Impact of Tariff and Non-Tariff Trade Barriers on Strategic Autonomy

Agenda 02: Strengthening Global Energy Supply Chain amidst Recent Geopolitical Developments

MUMBAIMUN'25

Letter from R&D

Dear Delegates,

It gives the Department of Research & Development, MUN Society MPSTME, immense pleasure to welcome you to the 12th edition of Mumbai MUN! True to our legacy, we remain committed to upholding and building upon a tradition of fostering informed discussions on matters of widespread concern.

We are primarily responsible for formulating relevant agendas and developing study guides that aim at imparting a holistic understanding of the issues at hand. This document has been prepared with a *first principles* approach, meant to play a part in orienting the committee discussions towards one that rises above the convention, and is characterised by substantiveness and inclusivity.

This year's theme, 'Voice Amidst the Void,' stands to remind us that when ideals fade into absence, voices must prevail. In an increasingly complex world, with an absence of conviction and shared purpose, we believe that the introduction of new ideas and impactful discourse is the need of the hour. True progress requires us to have the courage and willingness to speak up, have our perspectives heard, and to make sure that no voice is lost amidst a void of apathy.

Keeping this theme in mind, we have deliberately curated committees and agendas that focus on issues often overlooked in global conversations, in order to foster meaningful dialogue and resolution where it is most needed. In this study guide, you will find well-researched background information to help you prepare for your committee sessions. As facilitators, our aim is to equip you with the resources and insights you need to make the most out of your experience at Mumbai MUN.

We look forward to your active participation and contributions during Mumbai MUN and are confident that, together, we can make this conference an inspiring, enriching, and especially memorable experience for all.

Best wishes,
Department of Research & Development,
MUNSociety MPSTME.

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COMMITTEE OVERVIEW

The Group of Twenty (G20) was established in 1999 following the Asian Financial Crisis as a platform for Finance Ministers and National Bank Governors. It was later elevated to the level of Heads of State. Under a rotating presidency, the G20 Summit has been held annually since 2011. The committee initially focused on broad macroeconomic policy but has since expanded its scope to include trade, climate change, sustainable development, and energy. [1]

The G20's work is divided into the Finance and Sherpa Tracks. Within the two tracks, representatives from the relevant ministries of members and invitees, and various international organisations participate in working groups centred on a specific agenda, which is influenced by current economic developments as well as by the tasks and goals of the previous years. [2]

Guest countries and international organisations are also invited (with Spain being a permanent invitee). Regular participants include the World Bank, the United Nations, the World Trade Organisation, as well as the countries holding the presidency of regional organisations such as ASEAN and the African Union. Other countries may also be invited by the incumbent presidency.

MANDATE

The G20 is an international forum of developing and developed countries seeking to find solutions to global economic and financial problems. The G20 can be understood as a process rather than a set of discrete events, which concludes with an annual leaders' summit at which the participating heads of governments and states seek to agree on a communiqué setting out their agreements on key issues. The agreements are non-binding, and each of the participating heads implements most of the agreed points. Before the annual leaders' summit, working group meetings take place, which discuss agendas based on issue notes provided by the G20 Presidency. These issue notes include both unfinished prior issues and any new issues. The outcome of these meetings is reported, and disagreements are sought to be narrowed down so that chances of reaching an agreement are maximised. [3]

AGENDA OVERVIEW

Recent events, such as the Liberation Day Tariffs imposed by the United States, have significantly impacted global trade. Since the imposition of these tariffs, a debate about tariffs and non-tariff trade barriers has become imperative for countries to safeguard their economy and diversify their trade dependencies.

Trade plays a pivotal role in the pursuit of strategic autonomy for nations on the global stage. As the G20 represents 85% of global GDP and 75% of world trade. It has a fundamental responsibility to maintain economic stability and

[1] *What is the G20 and why does it matter?* World Economic Forum. (n.d.). https://www.weforum.org/stories/2024/11/g20-summit-what-you-need-to-know/

[2] Council on Foreign Relations. (n.d.-a). *What does the G20 do?* Council on Foreign Relations. https://www.cfr.org/backgrounder/what-does-g20-do

[3] Danny Bradlow Professor/Senior Research Fellow. (2025, July 22). *The G20: How it works, why it matters and what would be lost if it failed*. The Conversation. https://theconversation.com/the-g20-how-it-works-why-it-matters-and-what-would-be-lost-if-it-failed-251500

multinational cooperation to sustain world trade. This agenda explores the impacts of such trade barriers and discusses ways that a nation can maintain its strategic autonomy while preserving a harmonious global trade.[4]

BACKGROUND

Key Definitions

Tariffs

Tariffs are levied as taxes on products and services that are imported. As a result, indigenous goods are comparatively less expensive than imported ones.[5]

Non-Tariff Barriers

Regulations, guidelines, or practices that restrict imports or exports are known as Non-Tariff Barriers (NTBs). They are trade limitations rather than direct taxes.[6]

Trade Deficit

When a country imports more products and services than it exports, it has a negative trade balance. This is known as a Trade Deficit.[7]

Strategic Autonomy

The ability of a country to make choices independently without unduly depending on outside factors is known as Strategic Autonomy.[8]

Types of NTBs

Quotas

Commodities such as sugar are subject to quotas. They increase prices and restrict supply.[9]

Licensing Requirements

Include Pharmaceutical licenses. Administrative obstacles are increased by such regulations.

Technical Barriers to Trade (TBT)

Labelling regulations and safety requirements are examples. Despite their potential legitimacy, they are also employed in a protectionist manner.[10]

Currency Manipulation

Governments are attempting to alter the value of their currencies. This lowers the cost of a nation's exports for overseas consumers.[11]

Subsidies

Include those on US and EU domestic farms. Local products become artificially competitive due to these obstacles.[12]

Export Restrictions

Include prohibitions on the export of food. They affect international markets while preserving domestic supplies.

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4 *G20 members*. G20 South Africa. (n.d.). https://g20.org/about-g20/g20-members/
5 Hadden, R. (2025, July 31). *Tariffs 101: What are they and how do they work?*. Oxford Economics. https://www.oxfordeconomics.com/resource/tariffs-101-what-are-they-and-how-do-they-work/
6 *Non-Tariff Measures (NTMs)*. UN Trade and Development (UNCTAD). (2024, December 12). https://unctad.org/topic/trade-analysis/non-tariff-measures
7 *What is a Trade Deficit and how does it affect the economy?*. World Economic Forum. (n.d.). https://www.weforum.org/stories/2022/11/trade-deficit-global-economy/
8 *What is Strategic Autonomy?*. Telefónica. (2025, September 16). https://www.telefonica.com/en/communication-room/blog/what-is-strategic-autonomy/
9 *Understanding quotas: Trade restrictions explained*. Investopedia. https://www.investopedia.com/terms/q/quota.asp
10 *World Trade Organization*. WTO. (n.d.). https://www.wto.org/english/tratop_e/tbt_e/tbt_e.htm
11 Exchange rates, international trade and trade policies. (n.d.). https://unctad.org/system/files/official-document/itcdtab57_en.pdf
12 Distortive subsidies and their effects on Global Trade. (n.d.). https://thedocs.worldbank.org/en/doc/0534eca53121c137d3766a02320d0310-0430012022/related/Unfair-Advantage-Distortive-Subsidies-and-Their-Effects-on-Global-Trade-2023.pdf

Why do Countries impose Tariffs/NTBs?

Governments impose Tariffs or NTBs to protect the domestic industries and employment. Revenue can also be generated, and national security in delicate areas like food and defence can be maintained. These measures also allow addressing trade imbalances by being used as retaliation during conflicts or as regulations to safeguard public health and safety.[13]

Impact of Tariffs/NTBs on Strategic Autonomy

Trade barriers affect the ability of a country to independently make decisions in international affairs. The concept of Strategic Autonomy has multiple dimensions, which include: economic (reducing reliance on single suppliers), technological (building domestic capacity), defence (securing supply of sensitive goods), and political (maintaining sovereignty in decision-making). A country's autonomy may be restrained due to over-dependence on imports or foreign markets, hence emphasising the importance of trade policies in defining independence.[14]

World Trade Organisation (WTO)

The World Trade Organisation was established to succeed the General Agreement on Tariffs and Trade (GATT). Its functions include trade facilitation and global economic prosperity.[15] Also, disputes or grievances between members are resolved by the Appellate Body of the WTO. However, the United States stopped the process of reappointing judges after 2017, and in December 2019, the number of judges in the court fell below the minimum required number of three. The Appellate Body is now unable to review new applications, and there is great uncertainty over the WTO's dispute settlement process.[16] The prevalence of tariffs and NTBs has also increased.

HISTORICAL CONTEXT

Smoot-Hawley Tariff Act

The Smoot-Hawley Tariff Act of 1930 was implemented in the United States to protect American farmers and businesses from foreign competition. It increased tariffs by over 20% on imported goods. This had significant economic consequences as it triggered a retaliatory chain of tariffs from over 25 other nations.[17] International trade partners, like Canada and European countries, imposed high tariffs on American goods, reducing global trade.

This prolonged the Great Depression as exports of numerous countries were greatly impacted. The act not only had economic impacts but also had geopolitical consequences as international relations were dampened, which was evident when other nations targeted US exports with similar taxes. For instance, Canada placed tariffs on 16 goods, which cumulated to one-third of US exports, and European countries like France and Spain applied taxes on American automobiles.[18]

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[13] Hadden, R. (2025, July 31). *Tariffs 101: What are they and how do they work?*. Oxford Economics. https://www.oxfordeconomics.com/resource/tariffs-101-what-are-they-and-how-do-they-work/

[14] *Strategic Autonomy in Trade*. KPMG. (n.d.). https://assets.kpmg.com/content/dam/kpmgsites/sa/pdf/2025/strategic-autonomy-in-trade.pdf.coredownload.pdf

[15] *World Trade Organization*. WTO. (n.d.). https://www.wto.org/english/thewto_e/whatis_e/inbrief_e/inbr_e.htm

[16] *The World Trade Organization: The appellate body crisis: Economics program and Scholl Chair in international business*. CSIS. (n.d.). https://www.csis.org/programs/economics-program-and-scholl-chair-international-business/world-trade-organization

[17] Kenton, W. (n.d.). *What is the smoot-hawley tariff act? history, effect, and reaction*. Investopedia. https://www.investopedia.com/terms/s/smoot-hawley-tariff-act.asp

[18] ABC News Network. (n.d.). ABC News. https://abcnews.go.com/Business/smoot-hawley-tariffs-trump/story?id=116381286

Cold War Trade Barriers

The Cold War saw the use of tariffs and non-tariff barriers as a geopolitical weapon used by both the United States and the Soviet Union. They were used as tools to strengthen alliances, deny resources to the other bloc and signal diplomatic positions.

The Western Bloc, which included the US and its allies, used the General Agreement on Tariffs and Trade (GATT) to reduce tariffs and achieve trade liberalisation with the bloc, and created the Coordinating Committee for Multilateral Export Controls (CoCom) to control its export of strategic goods to the Eastern Bloc. On the other hand, the Eastern Bloc, which consisted of the Soviet Union and its allies, used the Council for Mutual Economic Assistance (COMECON) as a forum to foster trade and reduce dependency on the Western Bloc. [19]

Apart from tariffs, NTBs played a crucial part as they were used for strategic purposes. The US used export controls to stop the trade of high-tech goods to the Eastern Bloc.[20] On the other hand, the Eastern Bloc used import licensing, quotas and other administrative hurdles to restrict imports from the Western Bloc.[21] Trade between the blocs was severely restricted, equivalent to a 48% tariff, reducing global trade and resulting in losses to both sides.[22]

COVID-19 Trade Restrictions

The pandemic wrought several disruptive changes to the nature of global trade and the economy at large. Countries banned the export of essential medical supplies like vaccines, masks and medicines as they looked to prioritise domestic needs. These bans led to a shortage of critical supplies and increased the volatility in their prices, disproportionately more in developing nations.[23]

The pandemic also created numerous non-tariff barriers that hindered the trading of other goods. These included sanitary regulations, customs checks, and the shutdown of several ports. In addition to this, factory shutdowns, workforce shortages, and transportation bottlenecks (such as limited air cargo capacity) also impacted the global supply chain.

Countries understood the risk of over-reliance on a limited number of suppliers, viewing health security and supply chain resilience as a vital part of national security and autonomy, and moved towards supply chain diversification, reshoring and nearshoring. [24]

Past Frameworks

London Leaders' Declaration (2009)

G20 nations pledged to establish an anti-protectionist framework in response to the

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19 (PDF) "International Trade in the Cold War" in the cambridge history of international law - international law during the Cold War (draft). (n.d.-f). https://www.researchgate.net/publication/375451641_'International_Trade_in_the_Cold_War'_in_The_Cambridge_History_of_International_Law_-_International_Law_during_the_Cold_War_draft

20 The New York Times. (n.d.-k). U.S. ASSAILED FOR CONTROLS ON HIGH-TECH EXPORTS. https://www.nytimes.com/1985/09/23/business/us-assailed-for-controls-on-high-tech-exports.html

21 The Soviet-bloc foreign trade system Nicolas SPULBZR*. (n.d.-l). https://scholarship.law.duke.edu/cgi/viewcontent.cgi?article=2804&context=lcp

22 The economic consequences of geopolitical fragmentation: Evidence from the Cold War | CEPR. (n.d.-j). https://cepr.org/voxeu/columns/economic-consequences-geopolitical-fragmentation-evidence-cold-war

23 *Covid-19: Measures affecting trade in Goods*. WTO. (n.d.). https://www.wto.org/english/tratop_e/covid19_e/trade_related_goods_measure_e.htm

24 *Export restrictions do not help fight covid-19*. UN Trade and Development (UNCTAD). (2021, June 11). https://unctad.org/news/export-restrictions-do-not-help-fight-covid-19

Financial Crisis of 2008. This framework included a 'Standstill' on trade barriers and a 'Rollback' on the existing ones. These measures were aimed at a way to open trade and to allow economies to recover from the Financial Crisis of 2008.[25]

Various G20 Leaders' Declarations (2010–2018)

The commitment to anti-protectionist policies was constantly renewed, making it a staple of G20 communiques. The organisation repeatedly showed its support for the WTO, being a major forum for multilateral trade systems.[26]

Osaka Leaders' Declaration (2019)

The declaration reinforced commitment to open trade and emphasised regulatory cooperation as it implicitly recognised that domestic regulation can be a powerful non-tariff barrier for global trade. [27]

Riyadh Leaders' Declaration (2020)

The declaration strongly reiterated its commitment to open trade in light of the pandemic. There was a focus on ensuring a smooth trade of essential goods that was hampered by the surge of export restrictions.[28]

Bali Leaders' Declaration (2022)

Due to the economic fallout from the COVID-19 pandemic and the war in Ukraine, the declaration had an explicit focus on non-tariff barriers, especially on food and energy. The declaration highlighted the importance of not imposing trade restrictions on food inconsistently with the relevant WTO provisions in order to maintain food security. [29]

New Delhi Leaders' Declaration (2023)

Titled "Unlocking Trade for Growth", the declaration showed support for a transparent, rules-based, anti-discriminatory multilateral trading system. With reference to the pandemic, the declaration discouraged trade restrictions on essential goods in a time of crisis.[30]

Rio Leaders' Declaration (2024)

The recent declaration continued the trend of promoting transparency, supporting open trade and avoiding restrictive measures in multilateral trading systems and recognising the WTO to be at the centre of it. [31]

CASE STUDIES

AfCFTA (African Continental Free Trade Area)

The AfCFTA (African Continental Free Trade Area) is the largest free trade area (in terms of number of participating countries), bringing together 55 countries of the African Union and 8 Regional Economic Communities. It is aimed at eliminating trade barriers and promoting intra-African trade across all sectors of the African economy. Its objectives include having a single

[25] 2009 G20 London Summit Global Plan for Recovery and Reform. (n.d.-a). https://g20.utoronto.ca/2009/2009communique0402.html
[26] *2016 G20 trade ministers statement. (n.d.-b). http://www.g20.utoronto.ca/2016/160710-trade.html*
[27] *G20 Osaka Leaders' declaration: Documents and materials*. G20 Osaka Summit 2019. (n.d.). https://www.mofa.go.jp/policy/economy/g20_summit/osaka19/en/documents/final_g20_osaka_leaders_declaration.html
[28] G20 Riyadh Leaders' declaration. (n.d.-b). https://www.g20.utoronto.ca/2020/2020-g20-leaders-declaration-1121.html
[29] G20 Bali leaders' declaration Bali, Indonesia, 15-16 November 2022 1. (n.d.-b). https://g20.org/wp-content/uploads/2024/09/2022-11-16-g20-declaration-data.pdf
[30] G20 New Delhi Leaders' declaration New Delhi, India, 9-10 September 2023. (n.d.-c). https://www.mea.gov.in/Images/CPV/G20-New-Delhi-Leaders-Declaration.pdf
[31] G20 Rio de Janeiro Leaders' Declaration. (n.d.-c). https://g20.org/wp-content/uploads/2024/11/G20-Rio-de-Janeiro-Leaders-Declaration-EN.pdf

African market for trade, having common policies, making it easier for smaller businesses to trade, removing NTBs that hinder intra-African trade, lowering taxes and tariffs between member countries, and developing facilities that promote trade such as building roads.

The AfCFTA intends to create a comprehensive and mutually beneficial trade agreement amongst State Parties, covering trade in goods and services, investment, intellectual property rights, competition policy, digital trade, and women and youth in trade. Liberalising trade and progressively eliminating tariffs remains as their main objective.

Altogether, the AfCFTA includes a market of approximately 1.3 billion people and has a GDP of USD $3.4 trillion.

Tariff liberalisation became effective on 1 January, 2021 and followed the progressive elimination of duties as given below:

1. 90% coverage of tariff book (Category A) over a period of between five and 10 years.
2. 7% coverage of tariff book (Category B: sensitive products) over a period of between 10 and 13 years.
3. 3% coverage of tariff book (Category C: excluded products) exempt from tariff liberalisation.[32]

The AfCFTA pact has been signed by 54 of the 55 countries, but the implementation remains slow, with just 24 countries trading under it, according to AfCFTA Secretary-General Wamkele Mene. Experts argue that this is because of a variety of reasons, including prioritizing foreign trade agreements, economic disparity among countries, and dependence on foreign companies for shipping.[33]

India's Agricultural Tariffs

Some of the world's highest agricultural tariffs are imposed by India to protect its farmers. India has an average bound tariff of 113.1% on agricultural products, according to the WTO's 2024 Tariff Profile. Furthermore, the Most Favoured Nation (MFN) applied rates average around 36.7%.[34] Under this principle, a country is required to grant any other country the same trade advantages it grants to its "most-favoured" trading partner. As a result, the government can take actions such as raising additional tariffs in response to domestic pressures. India also maintains a Duty-Free Tariff Preference (DFTP) scheme for the Least Developed Countries (LDCs). Its objective is the grant of tariff preferences on the exports of the LDCs on imports to India. The scheme now provides for margins of preference or duty-free imports for about 98.2% of the imported products.[35]

However, India often faces criticism from trade partners for using such tactics. They assert that India's tariffs are unfair and violate legally binding agreements. Critics sometimes also argue that such practices can lead to an increase

32 AFCFTA – the Department of Trade Industry and Competition. (n.d.). https://www.thedtic.gov.za/sectors-and-services-2/1-4-2-trade-and-export/afcfta-2/
33 Grain. (n.d.). *Five years later, is the AFCFTA already failing? Business Africa*. Home. https://www.bilaterals.org/?five-years-later-is-the-afcfta
34 India part A.1 tariffs and imports: Summary and duty ranges total AG Non-Ag. (n.d.). https://www.wto.org/english/res_e/statis_e/daily_update_e/tariff_profiles/in_e.pdf
35 India's Duty-Free Tariff Preference (DFTP) Scheme for Least Developed Countries (LDCs). (n.d.). https://www.commerce.gov.in/17182-2/

in food prices and a reduction in competitiveness.[36]

Sanctions on Russia Post 2022

Since February 2022, when the Russian military invaded Ukraine, there have been several sanctions placed on Russia, led by the European Union and the G7 group of countries. These sanctions have had huge impacts on global trade as several nations have had to diversify their supply chains in order to reduce their dependence on Russia for certain goods. These include metals like aluminium and uranium; refined petroleum products; and natural gas. [37]

The EU has imposed various import and export restrictions on several products, resulting in a 61% decline in exports to Russia and an 89% drop in imports from Russia between the first quarter of 2022 and the second quarter of 2025, according to the data from Eurostat. U.S. imports from Russia fell to $2.50 billion in the first half of 2025 from $14.14 billion four years earlier, according to U.S. Census Bureau and U.S. Bureau of Economic Analysis data. Since January 2022, the United States has imported $24.51 billion of Russian goods.[38]

Liberation Day Tariffs

The 2nd of April, 2025, saw US President Donald Trump imposing import duties and tariffs on more than 90 countries, leading to a global market crash. He signed Executive Order 14257, which put a baseline of 10% tariff on all imported goods starting from 5th of April, with country-specific tariffs to begin from 9th of April.

As of 6th of August, the countries with the highest tariffs are India and Brazil at 50%. Trump aimed to impose these measures on India as a response to its purchases of Russian oil and weapons.[39] When it comes to Brazil, though, it is speculated that the high tariffs are influenced by politics.[40] Other countries with high tariffs, as of 6th of August, include Switzerland, Canada, Iran, China, South Africa and Mexico.[41]

President Trump also signed Executive Order 14256 to address the flow of illicit substances, specifically opioids from China into the United States. The order exempted the 'de minimis' rule, which means that all the imports from China and Hong Kong, irrespective of their prices, would be subject to duties and taxes. This order further increased the tensions between the two countries and fueled the trade war between them. [42]

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36 Chandra, A. (2021, December). *WTO rules against India's sugar export subsidies and domestic price support*. U.S. Department of Agriculture, Foreign Agricultural Service. https://apps.fas.usda.gov/newgainapi/api/Report/DownloadReportByFileName?fileName=WTO%20Rules%20Against%20India%27s%20Sugar%20Export%20Subsidies%20and%20Domestic%20Price%20Support_New%20Delhi_India_12-19-2021.pdf

37 *Import and export bans*. European Commission. (n.d.). https://commission.europa.eu/topics/eu-solidarity-ukraine/eu-sanctions-against-russia-following-invasion-ukraine/import-and-export-bans_en

38 US and Europe do billions in trade with Russia despite sanctions | Reuters. (n.d.). https://www.reuters.com/business/autos-transportation/us-and-europe-do-billions-trade-with-russia-despite-sanctions-2025-09-15/

39 Inamdar, N. (2025, August 27). *Donald Trump's 50% tariff on India kicks in as PM Modi urges self-reliance*. BBC News. https://www.bbc.com/news/articles/c5ykznn158qo

40 Wells, I. (2025, July 31). *Trump's Brazil tariffs are more about political revenge: Analysis*. BBC News. https://www.bbc.com/news/articles/cwy0147vxyqo

41 BBC. (2025, August 7). *Trump tariffs list: See all the tariffs by country*. BBC News. https://www.bbc.com/news/articles/c5ypxnnyg7jo

42 Federalregister. (n.d.-c). https://public-inspection.federalregister.gov/2025-06027.pdf?1743770717=

STAKEHOLDER ANALYSIS

Export vs. Import-Oriented States

Export-oriented nations see tariffs as barriers to growth, which reduce competitiveness in the global market[43], and also force retaliatory trade actions in some cases.[44] Examples of such countries include South Korea and Germany. China, despite being very Export-oriented, is one of the largest users of NTBs in the form of import restrictions, currency manipulation, and subsidies.[45] Its conflicts with the US over steel, technology, and agricultural products are examples of the risks that export-oriented economies face when trade barriers are escalated.

On the other hand, tariffs and NTBs are used as defensive measures by import-oriented states, which comprise mostly developing economies and countries with scarce resources. Some Western countries, too, have used trade restrictions and selective tariffs. The US imposes steel tariffs while EU members place carbon border taxes.[46] Nations in Sub-Saharan Africa and South Asia also impose tariffs. However, import dependence leaves them exposed to inflation. This may cause higher consumer prices and social unrest.[47]

Multilateral Institutions

A number of multilateral institutions are involved directly or indirectly in the decision-making and formulation of regulations. While the WTO is the official rule-maker, organisations like the United Nations Conference on Trade and Development (UNCTAD) also make a contribution. According to UNCTAD's analysis, NTBs currently have a greater impact on trade than tariffs and regularly prevent exports from developing countries.[48]

The World Bank (WB) provides financing, policy advice, and technical assistance to developing nations while the International Monetary Fund (IMF) helps in capacity development and discourages policies that would harm prosperity.[49] [50] Regional organisations like the African Union and the EU can function as a single bloc while negotiating trade barriers, or may be

43 *Trade Policy and the Global Economy*. OECD. (n.d.). https://www.oecd.org/content/dam/oecd/en/publications/reports/2018/08/trade-policy-and-the-global-economy-increasing-tariffs_d71b8be6/5dc62586-en.pdf
44 *Global Trade Outlook and Statistics April 2025*. WTO. (n.d.). https://www.wto.org/english/res_e/booksp_e/trade_outlook25_e.pdf
45 *China's use of unofficial trade barriers in the U.S.-China trade war*. FSI. (n.d.). https://sccei.fsi.stanford.edu/china-briefs/chinas-use-unofficial-trade-barriers-us-china-trade-war
46 *Carbon Border Adjustment Mechanism*. Taxation and Customs Union. (2025, August 29). https://taxation-customs.ec.europa.eu/carbon-border-adjustment-mechanism_en
47 *Sub-Saharan Africa: One Planet, Two Worlds, Three Stories*. IMF. (2021, October 21). https://www.imf.org/en/News/Articles/2021/10/20/pr21306-sub-saharan-africa-one-planet-two-worlds-three-stories
48 Insights from a new database the unseen impact of non-tariff measures: (n.d.). https://unctad.org/system/files/official-document/ditctab2018d2_en.pdf
49 *What We Do*. World Bank Group. (n.d.). https://www.worldbank.org/ext/en/what-we-do
50 *What is the IMF?*. IMF. (2024, October 21). https://www.imf.org/en/About/Factsheets/IMF-at-a-Glance

involved in establishing frameworks to eliminate internal tariffs. [51]

Strategic Economies

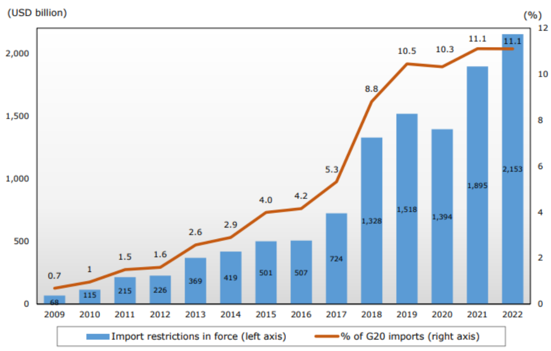
Global trade can, at times, be heavily influenced by a small group of countries. The US-China Tariff wars have reshaped the global supply chain, according to recent IMF work on geoeconomic fragmentation[52]. Resource-rich countries like Saudi Arabia and Brazil also impact trade. Exported products like oil and agricultural goods give them leverage. Strategically positioned countries like Singapore and Egypt, which control key chokepoints (such as the Strait of Malacca and the Suez Canal), also have an advantage. They can determine the flow of goods across critical routes.[53]

CURRENT CHALLENGES

Proliferation of NTBs

Non-tariff barriers are being applied increasingly to restrict trade measures and circumvent direct tariff impositions. The usage of non-tariff barriers is correlated with the income level of an economy.[54] High-income countries are more likely to use NTBs than low- or middle-income countries to protect the interests of their domestic industries and producers and to limit imports.

The proliferation of NTBs is straining diplomatic ties and can result in reciprocal tariffs. For example, the US imposed 25% tariffs on Indian goods with an additional penalty on Russian oil purchases and criticised New Delhi for its "strenuous and obnoxious" non-monetary trade barriers. The NTBs imposed by India included import caps, dairy certifications, and customs procedures, among others. [55]



Cumulative trade coverage of G20 import-restrictive measures on goods in force since 2009
Source: World Trade Organisation (WTO)

Supply Chain Fragmentation

A supply chain is a series of linkages from raw materials to the final product for end use and back to raw materials after use. Each stage of the supply chain involves a transformation step through a manufacturing or service process and a network of people, companies, organisations, and governments. Supply chain fragmentation refers to a system broken up into different parts, aiming to spread production across different suppliers and manufacturers across different countries.

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51 *AU in a Nutshell*. African Union. (2024, December 14). https://au.int/en/au-nutshell

52 MARIJN A. BOLHUIS is an economist in the IMF's Research Department., JIAQIAN CHEN is a deputy division chief in the IMF's Research Department., & BENJAMIN KETT is an economist in the IMF's Strategy. (2023, June 1). *The costs of geoeconomic fragmentation-Bolhuis-chen-Kett*. IMF. https://www.imf.org/en/Publications/fandd/issues/2023/06/the-costs-of-geoeconomic-fragmentation-bolhuis-chen-kett

53 World Oil Transit Chokepoints. International - U.S. Energy Information Administration (EIA). (n.d.). https://www.eia.gov/international/analysis/special-topics/World_Oil_Transit_Chokepoints

54 Non-tariff barriers and implications for International Trade. (n.d.-d). https://www.ifo.de/DocDL/ifo_Forschungsberichte_91_2017_Yalcin_etal_Protectionism.pdf

55 Key US complaints about India's "obnoxious" non-monetary trade barriers | reuters. (n.d.-h). https://www.reuters.com/business/finance/key-us-complaints-about-indias-obnoxious-non-monetary-trade-barriers-2025-07-31/

Since COVID-19, supply chain bottlenecks, such as the Ukraine conflict, labour shortages, and global instability, have disrupted the global supply chain. Supply chain disruption and trade distortions increase burdens for businesses and consumers, adversely affecting competitiveness and the cost of living. For manufacturers, increased global instability has caused commodity prices to rise to record levels and introduced significant cost volatility. [56]

The shift of manufacturing and exports moving to countries like India, Vietnam and Malaysia from China may seem like a healthy way to restructure the supply chain by diversifying it but, survey data indicate that for a substantial portion (17–34%) of manufacturing firms located in Germany, Italy, and Spain, China remains a key supplier of critical inputs that would be difficult to replace. Surveys also indicate that, beyond the trade channel, increased uncertainty is another important way in which rising geopolitical tensions impact firms. [57]

Price and Currency Fluctuations

The exchange rate of a country's currency is a significant determinant of its global economic status. Amidst the current trade disputes, an increase in tariffs can have two impacts. Firstly, import demand can be reduced by tariffs, causing a reduction in demand for foreign currency, as fewer businesses and consumers would be purchasing goods priced in that currency. This shift can lead to a depreciation of the exporting country's currency relative to the currency of the importing country. Secondly, tariffs might lead markets to expect tighter monetary policy to counter inflation, which could also cause the domestic currency to appreciate. Hence, some economists suggest that, apart from potentially making competitiveness suffer, the imposition of tariffs can also lead to an increase in domestic prices. However, countries mostly impose tariffs as a means to promote domestic employment or to mitigate the trade effects of foreign states. This may not always lead to price and currency fluctuations, but can facilitate them. [58]

Diplomatic Strains

Trade disputes have the potential to cause diplomatic ties to weaken and further exacerbate global tensions. An easy example to look at is the US-Brazil trade partnership. On 9 July 2025, the US imposed a 50% tariff on Brazil and then prompted an investigation to determine whether the Brazilian government acted in an unreasonable or discriminatory way against commerce with the United States. According to the Atlantic Council, a prolonged engagement between the two could lead Brazil to believe that the US is an unreliable trade partner, making them diversify their exports to other markets.[59]

Trade tensions have also eased between certain countries. On 9 September 2025, the US revised tariff rates on Japanese goods, striking a deal agreeing to reduce them to 15% in exchange for a $550 billion package of U.S.-bound investments and loans. There have been suggestions for similar agreements to be made to dismiss the

[56] *Supply chain bottlenecks: Causes, challenges, and resolutions*. Inbound Logistics. (n.d.). https://www.inboundlogistics.com/articles/supply-chain-bottlenecks-causes-challenges-and-resolutions/
[57] Trade wars and fragmentation: Insights from a new Escb report | cepr. (n.d.-g). https://cepr.org/voxeu/columns/trade-wars-and-fragmentation-insights-new-escb-report
[58] Ralph Ossa, C. E. (n.d.). *In a world of trade tensions, what do tariffs really do?*. WTO Blog. https://www.wto.org/english/blogs_e/ce_ralph_ossa_e/blog_ro_11apr25_e.htm
[59] Jcookson. (2025a, July 17). *The numbers that define the US-brazil trade partnership*. Atlantic Council. https://www.atlanticcouncil.org/blogs/new-atlanticist/the-numbers-that-define-the-us-brazil-trade-partnership/

diplomatic strains that have recently been formed.[60]

CONCLUSION

Trade barriers, both tariff and non-tariff, are used to ensure national security, shield domestic industries and manage trade imbalances. However, they can trigger retaliation, leading to geopolitical tensions and contraction in global trade. The current geopolitical situation- characterised by the proliferation of NTBs, like sanctions on Russia, and tariffs, such as the Liberation Day Tariffs- makes it imperative for G20, an organisation that represents 85% of global GDP and 75% of world trade[61], to provide a forum to address geopolitical tensions and preserve global trade.

[60] Japan says lower US tariffs will take effect by September 16 | reuters. (n.d.-g). https://www.reuters.com/business/autos-transportation/japan-says-lower-us-tariffs-will-take-effect-by-september-16-2025-09-08/

[61] *G20 members*. G20 South Africa. (n.d.). https://g20.org/about-g20/g20-members/

Agenda 2: Strengthening Global Energy Supply Chain amidst Recent Geopolitical Developments

AGENDA OVERVIEW

With the global energy consumption growing, it has become imperative that energy systems must be clean, efficient, and reliable. Since the Paris Agreement in 2015, clean energy sources have been promoted by many countries, which has further highlighted the importance of a stable energy supply chain.[62] Global disruptions have dictated energy availability, affecting energy security, incentivising countries to develop resilient supply chains.

This agenda explores the geopolitical and environmental challenges faced while strengthening global energy supply chains. Guidelines must be established to facilitate an affordable global energy network while ensuring sustainability and keeping in mind environmental concerns. Key issues that must be addressed include the shift to sustainable energy sources, the impact of Middle Eastern conflicts on energy affordability, and naval route discrepancies. By encouraging collaboration and innovation, the agenda seeks to ensure actionable policies that reinforce the global energy supply chain.

BACKGROUND

What is the Global Energy Supply Chain?

The global energy supply chain refers to the global system that extracts raw energy resources like oil, gas, or coal to convert them into usable forms like electricity or fuel and then delivers them to consumers across the world. It involves the process of generating energy and delivering it to consumers. Global energy supply chains represent the complex, worldwide system of processes and infrastructure required to bring energy resources from their origin to end consumers.[63]

What Constitutes the Global Energy Supply Chain?

The global energy supply chain relies on several steps, each of which needs to be executed perfectly for worldwide energy consumption. These steps include:

1. Exploration and extraction: This step includes identifying places from which raw energy resources can be extracted. After identification, energy plants are set up there to extract their resources.

2. Transportation: Next comes moving the resources across continents to distribution centres or consumers. Fossil fuels are usually transported through pipelines, ships, trucks, or railroads, while renewable sources of energy require specialized transportation methods.

3. Refining and processing: Raw energy resources must next be converted into usable forms, such as electricity, gas, or fuel. Fossil fuels need to be refined to remove impurities, whereas renewable energy sources need to be converted into electricity through turbines, solar panels, or batteries.

4. Distribution: This step involves delivering energy resources to consumers. Fossil fuels are usually distributed through gas stations or fuel depots, while electricity is distributed through grids or transmission lines.

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[62] Resilient supply chains for the Energy Transition. (n.d.-k). https://www.ecologic.eu/sites/default/files/publication/2023/30018-G20-Policy-Brief-Supply-Chains.pdf
63 Global Energy Supply Chains → term. Energy. (2025, March 2). https://energy.sustainability-directory.com/term/global-energy-supply-chains/

5. Consumption: This is the final stage of the energy supply chain, and involves the use of energy products by the end-users to power homes, operate factories, or fuel vehicles.[64]

Significance of Energy Supply Chain

Energy is quintessential to the global economy and disruptions in its supply may lead to energy losses, increased costs, and greater environmental impact.[65]

For example, the 2025 electric blackout in Spain caused widespread panic, crippling the workings of healthcare institutions, the failure of transport systems was a grievance, and the lack of cashless payments made buying daily essentials hard for many. A resilient energy supply chain reduces reliance on other countries, making it essential to national autonomy and is also instrumental for economic progress. Moreover, a stable energy supply chain facilitates the growth of renewable energy technologies, which are crucial for the energy transition for the betterment of the environment.[66]

Role of G20 in Energy Trade

G20 has specific bodies dedicated to confronting the rising energy costs and energy poverty. The G20 Energy Ministerial meetings and their Energy Declarations, along with the G20 Energy Sustainability Working Group (ESWG), focus on ways to improve global energy operations and act as guides for energy policies and international policies.

The annual Energy Declarations aim to provide a strategic framework for the global energy transition, having 4 main purposes: policy coordination between nations, investment in emerging green technologies, knowledge sharing between global organizations such as the International Energy Agency (IEA), and ensuring transitions are inclusive especially in energy-poor regions.[67]

HISTORICAL CONTEXT

Gulf War

The Gulf War in West Asia was triggered by Iraq's invasion of Kuwait, with the Iraqi leader Saddam Hussein, ordering it for various reasons, the most pressing being to acquire Kuwait's national oil reserves. The national revenue of Iraq depended mostly on oil, and Iraq considered it important to maintain high oil prices to ensure adequate revenue sources for the repayment of its debts and for its recovery from the Iran-Iraq war. Amidst this, crude oil prices dropped from $18 to $12 per barrel. Kuwait and the United Arab Emirates were blamed by Iraq for the price drop, claiming that it was caused by the overproduction of crude oil by those countries in disregard of the national production quota set for them by OPEC.

The price of oil before the Iraqi invasion was somewhere between $20-$40 per barrel, whereas, during the war, it peaked at $80-$90, dropping down to $37-$44 after the war ended. The output of oil produced by both countries had

64 Sarahrudge. (2024, May 17). *Navigating the five crucial stages of the energy supply chain: From exploration to consumption*. Energy, Oil & Gas magazine. https://energy-oil-gas.com/news/navigating-the-five-crucial-stages-of-the-energy-supply-chain-from-exploration-to-consumption/#:~:text=The%20supply%20chain%20of%20energy,and%20reliable%20supply%20of%20energy.
65 Global Energy Supply Chains → term. Energy. (2025a, March 2). https://energy.sustainability-directory.com/term/global-energy-supply-chains/
66 Guardian News and Media. (2025, April 30). *"no one knew what to do": Power cuts bring chaos, connection and Revaluation of Digital Dependency*. The Guardian. https://www.theguardian.com/world/2025/apr/30/no-one-knew-what-to-do-power-cuts-bring-chaos-connection-and-revaluation-of-digital-dependency
67 *AI-powered energy optimization. unlock cost savings.* nzero. (n.d.). https://nzero.com/blog/g20-energy-declaration-background-purpose-and-future-trends-explained/

reduced dramatically between 1990 and 1991, with some economists believing it to be the cause of the economic recession of 1991. The high oil prices caused by the war resulted in global economic instability. [68]

Fukushima Nuclear Disaster

The Fukushima Nuclear Disaster occurred on March 11, 2011, after a major tsunami had hit Japan due to an earthquake. The tsunami interrupted the power supply and the cooling factors of the nuclear power plant at Fukushima Daiichi, causing radioactive materials to leak and be released, leading to the death of about 2300 people. The event led to the efforts of the Japanese government to shift largely towards reducing its dependence on nuclear power and shut down all its nuclear facilities by 2014. Due to this, they had a 30% reduction in energy supply, which they filled via fossil fuels. In 2014, a Strategic Energy Plan (SEP) to further reduce the use of fossil fuels, increase the use of renewables, and restart nuclear power generation at the plants that were shut down was introduced.

Globally, countries like the USA, China, and Russia all responded to the event moderately, conducting surveys to check the safety of their plants and temporarily lowering their nuclear generation and replacing it with renewable sources. Germany, however, took drastic measures post-Fukushima, immediately deciding to phase out nuclear energy by 2022 and closing eight of its oldest nuclear power plants. They have also shifted to renewable energy and currently have no operating nuclear plants.

The immediate global response to Fukushima was most likely a "shock response," which temporarily slowed nuclear power generation in most places but also had drastic long-term impacts in other countries. The event caused a global fear of nuclear energy which eventually lessened. This fear also led to a measure increase in global renewable energy generation.

The disaster at Fukushima led to increased renewable energy development in Japan and Germany to become independent from nuclear power. More importantly, the immediate response of Japan closing down its nuclear power plants led to an increased demand for fossil fuels. The increased demand in Japan reduced global availability, increasing LNG prices, affecting global energy security for a short period of time, highlighting its delicacy. [69]

Abqaiq-Khurais Attack

The Abqaiq-Khurais attack refers to the 18 drones that hit the Abqaiq oil processing facility (the world's largest) and 4 missiles that struck the Khurais oil field, owned by Aramco, in Saudi Arabia. The Houthis, an armed Yemeni political and religious group, had claimed responsibility for the attack immediately after, while the US blamed Iran; however, there is no conclusive evidence that either is responsible for it.

At Abqaiq, the drones damaged 11 spheroid separation tanks used to separate gas from crude oil and stabilise it for export. Five stabilisation towers used to remove hydrogen sulphide from crude oil and two tanks that hold water removed from crude oil were also hit. At Khurais, four stabilisation towers were damaged. The Khurais oilfield produces about 1% of the world's oil, and Abqaiq has the capacity to process 7% of the global supply.

The Brent crude oil prices opened on Monday after the attacks at $71.57 per barrel (b) at 6:00 p.m. Eastern Standard Time (EST) on September 15, 2019, before falling to $67.00/b within 23 minutes of trading and $65.09/b after a further

[68] Ufheil-Somers, A. (2016, October 11). *Oil and the Gulf War*. MERIP. https://merip.org/1991/07/oil-and-the-gulf-war/
[69] The effect of the Fukushima nuclear disaster on the ... (n.d.-c). https://www.hks.harvard.edu/sites/default/files/centers/mrcbg/files/127_final.pdf

nine hours of trading. The price was back to $57/b quickly. This was mainly due to the surplus oil Saudi Arabia had and the US shale oil production being at an unprecedented high.

The attack exposed the fragility of the global energy supply chain, signifying the dependence, especially of Asian countries, on Saudi Arabian oil exports. The initial supply cut was of 5.7 million barrels per day, an oil disruption bigger than that of what was seen during the Gulf War. The event also brought to light that any conflict in the Gulf would carry huge risks for Saudi Arabia's oil infrastructure and the world economy. [70]

South China Sea Territorial Disputes

The South China Sea is one of the busiest maritime trade routes in the world, moving an estimated $5 trillion dollars[71] in goods each year serving as a commercial hotspot. It also has unexplored natural resources embedded in its seabed which could be exploited by the state which has a legitimate right over the region. Thus, many countries have overlapping claims of the Sea, historically making Southeast Asian territorial disputes rooted in disagreements over territorial control of the region. [72]

China claims the largest portion of the South China Sea in an area demarcated by its so-called "nine-dash line". The line comprises nine dashes that extend hundreds of miles south and east from its most southerly province of Hainan. They have also not specified if the claim includes only land territory or maritime space within it as well. China's nine-dash line claim encircles many oil- and gas-rich zones off the shores of key regional producers, including Vietnam, Malaysia, Indonesia, the Philippines, and Brunei. [73]

In 2012, China announced a column of nine new exploration blocks that closely track the nine-dash line claim which extended into Vietnam's 200 nautical-mile EEZ. In May 2014, a new deepwater drilling rig, belonging to the Chinese National Offshore Oil Company (CNOOC), entered Vietnamese-claimed waters for over a month of drilling, sparking violent demonstrations and attacks on perceived Chinese businesses in Vietnam.[74]

In 2024, China and the Philippines had a heated encounter where Chinese Coast Guard ships blasted two Philippine vessels by water canons in Scarborough Shoal, an area claimed by China under the nine-dash line but coming under the Phillipine's EEZ.[75] This is just one of the many disputes that have arisen in recent years due to China's ambitious plans, the Philippines' new era of defiance, and members of ASEAN being divided over taking stances on the issue, fearing economic repercussions from China. These geopolitical developments are now frequenting

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70 Risk of oil supply disruptions can have an immediate effect on oil prices - U.S. Energy Information Administration (EIA). (n.d.).
https://www.eia.gov/todayinenergy/detail.php?id=42675

71 Schrag, J. (2021, January 25). *How much trade transits the South China Sea?*. ChinaPower Project.
https://chinapower.csis.org/much-trade-transits-south-china-sea/

72 Maritime chessboard: The geopolitical dynamics of the south china sea | geopolitical monitor. (n.d.-i).
https://www.geopoliticalmonitor.com/maritime-chessboard-the-geopolitical-dynamics-of-the-south-china-sea/

73 BBC. (2023, July 7). *What is the south china sea dispute?*. BBC News. https://www.bbc.com/news/world-asia-pacific-13748349

74 *The role of energy in disputes over the South China Sea*. The Role of Energy in Disputes over the South China Sea | The National Bureau of Asian Research (NBR). (n.d.). https://www.nbr.org/publication/the-role-of-energy-in-disputes-over-the-south-china-sea/

75 Al Jazeera. (2024, May 1). *Philippines and China in new confrontation at Scarborough shoal*.
https://www.aljazeera.com/news/2024/4/30/philippines-and-china-in-new-confrontation-at-scarborough-shoal

the encounters in the region while also making them more confrontational. [76]

Past Frameworks

G20 Hamburg Climate and Energy Action Plan

In 2017, G20 (with the exception of the US) decided on clear measures for implementing the Paris Agreement and commencing the global energy transition in line with the goals of the 2030 Agenda for Sustainable Development.[77]

2022 G20 Bali roadmap

The Bali Roadmap intends to focus on a set of core priorities, milestones as well as voluntary and desired collaborative actions that will, naturally, reflect national circumstances, needs and priorities. It prioritizes energy security, scaling up smart and clean energy sources, and financing clean energy initiatives. [78]

G20 Indian Presidency

The G20 Leaders' Declaration stressed on the need for a diversified, sustainable, and responsible supply chain for energy transition, including for critical minerals.[79]

ALIGNMENT WITH SDGS

The 2030 Agenda for Sustainable Development, adopted by all United Nations Member States in 2015, provides a shared blueprint for peace and prosperity for people and the planet, now and into the future. At its heart are the 17 Sustainable Development Goals (SDGs), which are an urgent call for action by all countries - developed and developing - in a global partnership[80].

SDG 7, which is Affordable and Clean Energy, promotes the use of renewable energy while making sure that everyone gets access to it. Its goal is to "Ensure access to affordable, reliable, sustainable and modern energy for all." This is directly aligned with the Global Energy Supply Chain. Other than this, SDG 9, SDG 12, SDG 13 and SDG 17 are vital aspects when we talk about renewable energy as they promote sustainability and taking responsibility.

RECENT GEOPOLITICAL DEVELOPMENTS

Russian-Ukrainian War

[76] Bautista, D. L. (n.d.). *Rising tensions in the South China Sea: The strategic calculations at play*. Australian Institute of International Affairs. https://www.internationalaffairs.org.au/australianoutlook/rising-tensions-in-the-south-china-sea-the-strategic-calculations-at-play/
[77] Bmukn. (n.d.). *G20 Hamburg climate and Energy Action Plan for Growth - key points and measures*. Federal Ministry for the Environment, Climate Action, Nature Conservation and Nuclear Safety. https://www.bundesumweltministerium.de/en/topics/international/overview-international/g7-and-g20/germanys-g20-presidency/g20-hamburg-climate-and-energy-action-plan-for-growth-key-points-and-measures#:~:text=With%20the%20G20%20Hamburg%20Climate,2030%20Agenda%20for%20Sustainable%20Development.
[78] Bali Energy Transitions Roadmap. (2022). https://www.g20.utoronto.ca/2022/Bali-Energy-Transitions-Roadmap_FINAL_Cover.pdf
[79] G20 New Delhi Leaders' declaration New Delhi, India, 9-10 September 2023. (n.d.-c). https://www.mea.gov.in/Images/CPV/G20-New-Delhi-Leaders-Declaration.pdf
[80] United Nations. (n.d.). *Transforming our world: The 2030 agenda for sustainable development | department of economic and social affairs*. United Nations. https://sdgs.un.org/2030agenda

Russia-Ukraine Gas War
Source: Getty Images

The ongoing Russian-Ukrainian War has had several impacts on the global energy supply chain. According to the IEA, this war triggered the first truly global energy crisis, and its effects can still be seen in many countries[81]. The sanctions imposed on Russia's exports of oil, natural gas, LNG, and other energy sources by the EU and the G7 countries resulted in higher energy prices due to an energy shortage.

Moreover, there was a huge shift in energy supply sources for the EU and the G7 countries.[82] They looked to countries like the United States and Norway for LNG and natural gas supplies. However, Russian exports remained large as they started exporting more to other countries like China, India, Türkiye and some in the Middle East.

The war has also exposed the fragility of energy infrastructure on the continent after several underwater explosions occurred near the Nord Stream. These pipelines were used to transport natural gas from Russia to Germany through the Baltic Sea. Even though the pipelines were not transporting energy due to the ongoing conflict, they were filled with natural gas. The subsequent gas leaks led to 456,000 tons of methane gas escaping.[83] This not only had an impact on the global energy supply chain, as the gas prices rose by 19%, but it also had a detrimental impact on the environment, as it is the largest single methane leak in history.[84]

Middle Eastern Tensions

The past few years have seen rising tensions in the Middle East, with several countries declaring war on each other and engaging in conflicts. These pose a significant threat to maritime chokepoints, such as the Suez Canal, Strait of Hormuz, and Bab al-Mandab Strait. Ships have been trying to avoid these chokepoints by going around the African continent, which raises the voyage time, freight rates and fuel use, impacting global trade. These longer routes have resulted in increased congestion at ports, higher fuel consumption, higher crew wages and insurance premiums, and increased exposure to piracy. Global ton-miles grew by 4.2% in 2023, driving up costs and emissions.

Moreover, the conflicts have triggered sharp fluctuations in oil prices. The price of Brent crude, the global benchmark for oil, surged to a five-month high of $81.40 following the bombing of nuclear sites in Iran by the US. However, news of a ceasefire has helped the price to fall back to approximately $69 a barrel. Oil prices are closely monitored, as they can have far-reaching consequences for countries worldwide. Rising costs of energy can make many goods more expensive, as seen following Russia's invasion of Ukraine three years ago. The conflict also raised concerns that Iran could decide to block traffic going through the Strait of Hormuz, one of the

[81] Russia's war on Ukraine – topics - IEA. (n.d.-f). https://www.iea.org/topics/russias-war-on-ukraine
[82] *Statistics explained*. EU imports of energy products - latest developments - Statistics Explained - Eurostat. (n.d.). https://ec.europa.eu/eurostat/statistics-explained/index.php?title=EU_imports_of_energy_products_-_latest_developments
[83] Energynews. (2025, January 17). *Sabotage of nord stream pipelines: Political and economic impact of record methane emissions - energynews*. energynews.pro. https://energynews.pro/en/sabotage-of-nord-stream-pipelines-political-and-economic-impact-of-record-methane-emissions/
[84] Fleming, S. (2022, October 12). *International impacts of the Nord Stream Leaks*. globalEDGE Blog: International Impacts of the Nord Stream Leaks >> globalEDGE: Your source for Global Business Knowledge. https://globaledge.msu.edu/blog/post/57169/international-impacts-of-the-nord-stream

world's most important oil shipping routes, as about 20% of global oil and gas flows through it.[85]

The conflicts in the Middle East and the maritime chokepoints present there have cast a shadow of uncertainty on the global energy supply chain.

Pirates and Mercenaries

Maritime piracy has increased in recent years, becoming a global issue as it is no longer linked only to Somalia, spreading to other areas like the Gulf of Guinea and the Singapore Straits. They have cost $18 billion to the global economy, according to the World Bank. Pirates use violence to illegally board, loot and seize vessels, which not only threatens global trade but also the lives of crew members.[86]

Several mercenary groups can also apply pressure on International Trade. One example is the group named Ansar Allah, also known as Houthis. Houthis have led to several attacks in the Red Sea, making it a maritime chokepoint. These attacks have led to a drop of average daily trade volume from 4.89 million metric tonnes to 1.36 million metric tonnes within the last year.[87]

Constant attacks force shipping companies to take longer routes to avoid the high-risk areas, which increases the freight prices and overall increases the price of goods like oil and gas.

Post COVID-19 Disruptions

The pandemic led to a significant decline in global energy demand. Even though residential electricity demands increased due to lockdowns, this was outweighed by the huge decline in commercial and industrial energy demands. This helped countries to accelerate towards decarbonisation and to build renewable energy sources, establishing energy security and resilience.

Global energy markets have stabilised through the coordinated efforts of energy-producing countries in the OPEC-plus group, as well as closer collaboration between G20 and IEF member countries.[88] G20's primary focus was on fostering energy security and reinstating market stability during the time.

However, the crisis induced by the pandemic has raised several new challenges about extreme energy market volatility and the need to understand how market mechanisms and fundamentals drive energy markets forward. Dialogue and collaboration on energy market security and data transparency, as well as available spare capacity, inventories, and collective emergency response and prevention measures, have grown more crucial than ever.

CASE STUDIES

India's Oil Diversification

India imported 232.5 MMT of crude oil in the financial year 2023-24. This accounts for more than 85% of its total oil consumption[89]. India has diversified its oil sources to meet demands and to keep avenues open in case of price swings and

[85] *Vulnerability of supply chains exposed as global maritime chokepoints come under pressure*. UN Trade and Development (UNCTAD). (2024, October 22). https://unctad.org/news/vulnerability-supply-chains-exposed-global-maritime-chokepoints-come-under-pressure

[86] Admin. (2024, May 2). *The impact of maritime piracy on the global economy*. Onus Insurance. https://onusinsurance.com/the-impact-of-maritime-piracy-on-the-global-economy/

[87] Power, J. (2024, October 6). *Houthi Red Sea attacks still torment global trade, a year after October 7*. Al Jazeera. https://www.aljazeera.com/economy/2024/10/5/a-year-after-october-7-houthi-red-sea-attacks-still-torment-global-trade

[88] *Energy security and stable markets after covid-19: IEF*. International Energy Forum. (n.d.-b). https://www.ief.org/focus/energy-security/energy-security-stable-markets-after-covid-19

[89] *Oil for India*. Oil for India | The National Bureau of Asian Research (NBR). (n.d.). https://www.nbr.org/publication/oil-for-india/

regional instability. Earlier, India sourced oil primarily from the Middle East, with almost none of its oil supply coming from Russia. Since 2022, however, India has increased its oil purchases from Russia exponentially, despite Western sanctions.

New Delhi has also broadened its imports from countries in Africa, North America, and Latin America. In recent times, India has increased its buying of US oil and LNG, drawn by competitive prices. African nations like Nigeria, Angola and Algeria are also top suppliers of crude oil to the country. At the same time, investments in strategic reserves, refining capacity and renewable alternatives are also promoted.[90]

China's Rare Earth Metals Dominance

Rare earth metals consist of a group of 17 elements, namely the lanthanides and scandium and yttrium, which are used in various consumer products such as EVs, aircraft engines, medical equipment, oil refining, and missile and radar systems. They are critical elements in renewable energy technologies due to their unique chemical characteristics. China has rare earth element deposits concentrated in southern China. They realised the importance of these metals in the 1960s, and with help from the US, which had the biggest market share of rare earths during the 20th century, they improved on existing technologies and started lucrative mining and processing firms with the help of the cheap electricity available. In 2024, China controlled over 69% of rare earth production in the world and nearly half the mines.

China now hardly ever sells raw materials: it uses them to manufacture products with higher added value, such as magnets and electric motors. In 2019, China accounted for 92% of the world's production of rare-earth permanent magnets. This creates a monopoly for them and allows them to control prices and restrict the availability of these metals to certain countries. [91]

Japan has proposed a plan to the G7 to find and finance rare-earth mining and refining operations in Africa and Latin America in an attempt to take away the global reliance on these minerals from China and circumvent the trade restrictions China puts on countries regarding rare-earth minerals and take away the monopoly from them.[92]

Panama Canal Drought

The Panama Canal is a highly strategic 51 mile waterway connecting the Atlantic and Pacific oceans. It cuts down shipping times dramatically and accounts for 6% of world-wide maritime trade., handling $270 billion in cargo annually. It has been facing a drought since 2023 due to the lack of rain and the El Nino phenomenon.[93]

Shipments of two million tons of goods from textiles to food have been delayed because of the disruptions at the Panama Canal. The extended dry season in 2024 led to the number of transits per day being reduced to 32 in September and to

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90 India's crude oil imports remain stable at 232.5 million tons in FY 2023-24 | Reuters. (n.d.). https://www.reuters.com/business/energy/indias-crude-oil-imports-remain-stable-2325-million-tons-fy-2023-24-govt-data-2024-04-17/

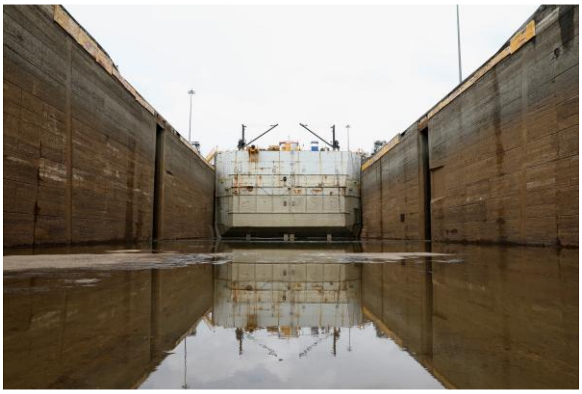
91 Caliman, L. (2025, February 2). *China has a monopoly on rare earth metals*. Polytechnique Insights. https://www.polytechnique-insights.com/en/columns/geopolitics/china-has-a-monopoly-on-rare-earths/

92 Ezrati, M. (2024, May 8). *How much control does China have over rare earth elements?*. Forbes. https://www.forbes.com/sites/miltonezrati/2023/12/11/how-much-control-does-china-have-over-rare-earth-elements/

93 Fleury, M. (2024, March 6). *Can the Panama Canal save itself?*. BBC News. https://www.bbc.com/news/business-68467529

24 in November, rather than the typical 36, which in turn caused significant delays. As fewer ships pass through the Canal, the competition for slots has driven up prices. Industries relying on imports are facing higher shipping prices due to reduced capacity. Supply chains are also being disrupted due to ships being rerouted, increasing delivery times.[94]

The Panama Canal is important for trade between the United States and East Asia and the western coast of South America. There is increased demand for ship traffic through the Panama Canal due to higher demand for U.S. propane in East Asia as a petrochemical feedstock. This demand is also contributing to delaying and making the route a larger chokepoint. The delays caused at the Canal have pushed shipping rates higher elsewhere by decreasing the globally available number of vessels as more wait to cross it.[95]



A view of the dry West Lane of Pedro Miguel Locks at the Panama Canal
Source: Reuters

STAKEHOLDER ANALYSIS

Producer and Consumer States

Energy-producing states are constantly trying to address supply chain bottlenecks and shift to renewable energy sources. Both of these options will prove to be economical to their economies in the long run, but come with considerable challenges. Global tensions and unsafe trade routes further worsen the bottlenecks, and the lack of rare-earth metals in the global market toughens the shift to renewable energy.

Consumer energy states, on the other hand, face rising energy prices caused by supply chain disruptions. This further worsens the cause of energy poverty in the world and slows the shift towards clean energy.

A country like Australia can be considered as both a consumer and a producer state. They rank second in the world as coal exporters and rank fifth in the Asia Pacific region as oil product importers. They play an important role in the total global energy supply, being the leading producers of oil, and are heavily dependent on refined oil product imports.[96]

India, on the other hand, is majorly a consumer state, being the third largest importer of crude oil in the world, second largest importer of coal, and thirteenth largest importer of natural gas. They are highly dependent on the smooth working of the global energy trade for a constant energy supply.[97]

[94] Jovanovic, D. (2025, June 12). *How panama canal disruptions are shaping global supply chains · log-hub*. Log. https://log-hub.com/how-panama-canal-disruptions-are-shaping-global-supply-chains/

[95] Panama Canal drought could threaten supply chain, S&P says | Reuters. (n.d.-c). https://www.reuters.com/business/environment/panama-canal-drought-could-threaten-supply-chain-sp-says-2024-04-03/

[96] Australia - countries & regions - IEA. (n.d.-b). https://www.iea.org/countries/australia

[97] India - Countries & Regions - IEA. (n.d.-e). https://www.iea.org/countries/india

A producer such as Saudi Arabia, which is the largest exporter of crude oil in the world, has negligible imports of coal and natural gas due to their overproduction of crude oil. Their oil export makes up a considerable amount of their GDP. A large portion of their revenue depends on their crude oil export, making them work towards fewer supply chain bottlenecks. [98]

Multilateral Institutions

The IEA tries to support all sources of energy, trying to ensure energy availability to everyone. They are ensuring oil, electricity, gas, and critical mineral security to their member-states. With threats to energy systems always evolving, the IEA monitors and analyses these threats and tries to prevent them from having a major global impact. The IEA also regularly works with the G20. At the third ETWG meeting in Sun City, a workshop was co-organised on G20 regional interaction and regulatory cooperation. [99]

An institution like OPEC solely works to stabilize oil markets, ensuring a safe and regular supply of oil to its consumers and a steady income for its members. OPEC also significantly influences crude oil prices by setting production goals for its member countries. After the imbalances caused in the energy market by the pandemic, OPEC worked closely with the energy-producing and consuming states of the G20, restoring energy market stability.[100]

An institution that directly examines the supply chain and addresses its challenges is the UNCTAD. It promotes mineral-rich developing countries in adding more local value and also advocates for a more traceable, trackable, and accountable energy supply chain which has reduced barriers in the market. It also promotes a clean energy transition to renewable sources. Under the Indonesian presidency of the G20, a compendium was prepared by UNCTAD which highlighted the efforts made by the G20 states to equip sustainable energy alternatives. [101]

CURRENT CHALLENGES

Geopolitical Rivalries

Countries that can potentially be very good trade partners sometimes end up competing with each other for economic, military and technological dominance. The Russia-Ukraine war has added to supply chain woes, particularly in African and EU countries, disrupting trade routes, vital supplies and operations. With oil and gas prices soaring, transportation costs have followed suit, with the war-imposed constraints on the ability to use Russian transportation infrastructure (such as the railways) to support manufacturing in Asia. Ukraine supplies about 50% of the world's semiconductor-grade neon gas[102], affecting secondary supply chains too. Moreover, both countries export grains as well as fertilisers.

Middle-Eastern Tensions have caused fuel costs to rise. Shipping routes have become increasingly unpredictable, with carriers choosing safer alternatives (even though they add weeks to shipments), such as the Cape of Good Hope, to

[98] Saudi Arabia - countries & regions - IEA. (n.d.-h). https://www.iea.org/countries/saudi-arabia
[99] IEA contributions to the G20 in 2025 - event - IEA. (n.d.-e). https://www.iea.org/events/iea-contributions-to-the-g20-in-2025
[100] *Speeches*. Organization of the Petroleum Exporting Countries. (n.d.). https://www.opec.org/speech-detail/355-2-september-2022.html

[101] *G20 compendium on promoting investment for Sustainable Development: (bali compendium)*. UN Trade and Development (UNCTAD). (2022, September 30). https://unctad.org/publication/g20-compendium-promoting-investment-sustainable-development
[102] Simchi-Levi, D., & Haren, P. (2022, March 17). *How the war in Ukraine is further disrupting global supply chains*. Harvard Business Review. https://hbr.org/2022/03/how-the-war-in-ukraine-is-further-disrupting-global-supply-chains

those under pressure from attacks and regional instability, including the Persian Gulf and the Suez Canal.

Balancing Climate Goals with Energy Security

Continued use of fossil fuels is seen throughout the world despite climate pledges. Governments plan to produce around 110% more fossil fuels in 2030 than would be consistent with limiting global warming to 1.5°C[103]. They have still not agreed to address the production of all fossil fuels. Without supportive policies and regulations, the widespread acceptance of renewable energy is hindered. Dependence on fossil fuels remains a challenge in shifting to renewable energy from the fossil fuel industry.

Countries such as Canada, Japan, and New Zealand have set net-zero targets for around 2050[104] (nearly three decades from now). However, long-term objectives require near-term action today. Although net-zero targets continue to gain traction with governments and companies, sceptical voices have emerged, saying that the 'net' aspect of net-zero targets could dampen efforts to rapidly cut emissions. Critics are concerned that some countries' net-zero targets rely on purchasing emissions reductions, delaying reductions within their own boundaries.

Unequal Access to Energy

There exists a great disparity in global energy access. A report by the WHO finds that the world remains off course to achieve SDG 7 for energy by 2030[105]. The number of people without access to electricity increased for the first time in over a decade, mostly in Sub-Saharan Africa, at a higher rate than that of new electricity connections. A combination of factors contributed to this, including inflation and growing debt distress in many low-income countries.

While people in the Global South need access to electricity for the most important basic human needs, for most people in the Global North, electricity and heat are available around the clock and for almost all needs.[106] However, it is not solely the geographical location that is decisive for access to energy; the socio-economic framework conditions are also significant.[107] Not everyone has sufficient financial means to be able to afford access to sufficient energy. It also becomes evident that not only in the Global South, but also in the Global North, discussions about affordable and sustainable energy for everyone must be addressed.

Infrastructural Vulnerabilities

Threats to the global energy systems continue to evolve. Cascade effects and Interdependencies can disrupt essential services, even when a single component fails. Natural disasters also affect

[103] Governments plan to produce double the fossil fuels in 2030 than the 1.5°C warming limit allows. (n.d.). https://www.unep.org/news-and-stories/press-release/governments-plan-produce-double-fossil-fuels-2030-15degc-warming

[104] *Senator Rosa Galvez*. List of Countries with Net-Zero Commitments and Climate Accountability Legislation | Senator Rosa Galvez. (n.d.). https://rosagalvez.ca/en/initiatives/climate-accountability/list-of-countries-with-net-zero-commitments-and-climate-accountability-legislation/

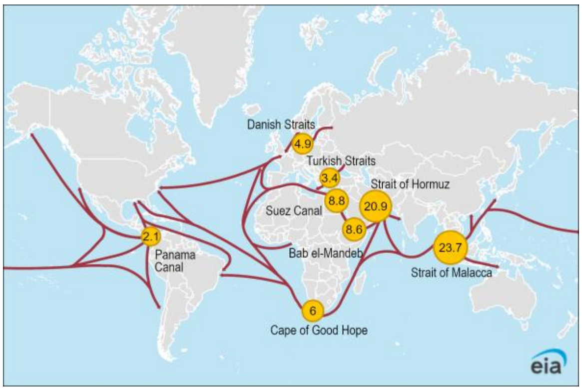
[105] World Health Organization. (n.d.). *Progress on basic energy access reverses for the first time in a decade*. World Health Organization. https://www.who.int/news/item/12-06-2024-progress-on-basic-energy-access-reverses-for-first-time-in-a-decade

[106] Secure and People-Centred Energy Transitions - World Energy Outlook 2023 - Analysis - IEA. (n.d.). https://www.iea.org/reports/world-energy-outlook-2023/secure-and-people-centred-energy-transitions

[107] United Nations. (n.d.). *SDG indicators - Goal 07*. https://unstats.un.org/sdgs/report/2024/goal-07/

energy trade. The 2011 earthquake and tsunami that damaged the Fukushima reactors led Japan to increase its LNG imports.

Cyberattacks have also become frequent. Vulnerabilities have been revealed by attacks such as those on Saudi Aramco in 2017 and ransomware incidents targeting US gas pipelines. Maritime chokepoints cause bottlenecks, too. An example can be Iran's seizures of tankers in 2019 near the Strait of Hormuz.



Global Oil Transit Routes
Source: U.S. Energy Information Administration

Market Volatility

Fluctuations in energy markets affect economic activities. Trade can be hindered by shipping delays and increased prices, which can reduce energy supply.

The requirements for continuity and stability of energy supply have significantly increased. However, the energy supply chain is often disturbed by the imbalance between supply and demand. For instance, in recent years, influenced by the COVID-19 pandemic, the vulnerability of energy supply chains has become more prominent.[108]

In many countries, energy commodity prices fluctuate sharply as the energy supply chain is disrupted. For instance, in some small island states, such as Oceania Island countries, affected by the characteristics of limited area, remote locations, and non-interconnected power networks, these countries are highly dependent on imported fossil fuels.[109] Thus, the fluctuation of energy prices and the interruption of the supply chain pose risks to their energy supply. Currency Fluctuations also contribute to volatility. As energy is largely priced in US dollars, it makes imports costlier for countries with weak currencies.

CONCLUSION

The global energy supply chain is highly fragile and is adversely affected by various economic, environmental, and geopolitical factors. According to a UNDP paper, 1.18 billion people are living in energy poverty.[110] A stronger energy supply chain free from bottlenecks and geopolitical disruptions can reduce energy disparity and also pave the way to an easier energy transition. There has not been a global united push for addressing the modern problems of the energy supply chain, and the fragmented efforts have been made in self-interest. A global effort is now needed to have accessible, affordable, and clean energy for everyone in the future.

108 *The Impact of COVID-19 on Energy Market Stability*. International Energy Forum (IEF). (n.d.). https://www.ief.org/_resources/files/news/analysis-reports/ief-insight-brief_the-impact-of-covid-19-on-energy-market-stability.pdf

109 *Islands tap sun and wind power as sea levels rise*. World Bank. (2022, September 25). https://www.worldbank.org/en/news/feature/2022/09/20/islands-tap-sun-wind-power-as-sea-levels-rise

110 *Beyond access: 1.18 billion in Energy Poverty...: Data Futures Exchange*. UNDP. (n.d.). https://data.undp.org/blog/1-18-billion-around-the-world-in-energy-poverty